

How does Linux differ from UNIX?

During the time from 1991 to 1995, many computing or engineering students were accustomed to the power of UNIX and X, at university. Many students had wonderful new things like E-Mail, the web and more. At home, they would have to make do with their 16 bit computers, waiting for all these powers to come to their homes one day. A lot of them felt like all this good stuff was being reserved for the computers at Uni. They had to make do with MS-DOS or Windows 3.1. UNIX was a big, expensive beast.

Enter: Linus Torvalds

Linus was, in 1991, a student in Finland studying Computer programming at The University of Helsinki. Linus used UNIX at University on a daily basis. He got bored of his 386 PC running MS-DOS, and decided to start his own UNIX kernel. A kernel is the name for the program at the heart of every operating system that talks to the hardware directly. He wanted to distribute the kernel software freely, because it was a hobby, not a commercial product. He also wanted to see what others thought of his work so he decided to publish it for free on the Internet. He finished the first Linux kernel in late 1991. In quite a short space of time he had made a 32 bit kernel, in which programs could be run. At that point he had a kernel, but he didn't really have any useful software to run on it.

Here is Linus' very first post to the Internet in 1991, about the creation of what would become Linux:

Hello everybody out there using minix –

I'm doing a (free) operating system (just a hobby, won't be big and professional like gnu) for 386(486) AT clones. This has been brewing since April, and is starting to get ready.

The Free Software Foundation and the GPL

Over in the USA, a fellow by the name of Richard Stallman had created a team (of programmers) devoted to free software. Richard called this the "[Free Software Foundation](#)". The goal of his organisation was to make software free to distribute, and free to obtain the source code along with it so that others could make improvements to the software through the Internet. The FSF created a bunch of tools that mirrored the functionality of their commercial, proprietary UNIX counterparts. These tools were called the GNU tools. GNU stands for 'GNU's Not Unix'. Don't ask!

The Free Software Foundation needed a license for the software that they (and anyone else) could use it. The license was based on the principles of CopyLeft. It was released as the GNU GPL (General Public License). The GPL is now at version 3, but at the heart of it, the license still has the same pretenses: That any authors of the software using the GPL could charge for the software, as long as they are willing for it to be freely distributed. By creating community-based software, that has open standards and is subject to peer review, the quality of the software would be good. Richard Stallman may be an eccentric hippie, but he gave birth to the whole wonderful world of open source, from as far back as 1984.

Stallman had been busy making a whole suite of software, for example: a text editor called emacs, and bash (the Bourne Again Shell) which is a command line interface based upon the original Bourne Shell that comes with the BSD variant of UNIX. The FSF's software was entirely based upon the UNIX software suite, and generally improved on it. In 1991, the only thing that the FSF were missing to make it a fully fledged operating system was The Kernel (the brains of the operating system).

Linus managed to ensure that his kernel would work with the GNU's apps. Once Linus made the combination of the GNU UNIX applications, the code compiler (gcc) and a working kernel, Linus had something that worked! Linus decided to call his creation Linux.

Linux is pronounced 'Lih-nucks' not 'Ly-nucks'. Here's Linus Torvals pronouncing it!

Now, over twenty five years on from the original post on the Internet, Linus is still working on Linux, but it's now an effort which is collaboratively worked upon by millions of other individuals, corporations and organisations around the world. You probably don't know it, but Linux is everywhere today. It's on your Android mobile phone, in your TV and nine out of ten websites run Linux on their servers (think Google, Amazon and Facebook to name but a few), Linux might even be in your new dishwasher or fridge-freezer!

Now, visit [Chapter Two](#) to see why using Linux instead of Microsoft Windows or MacOS may be the best thing you ever did with your computer!